Discourse status as a cue to the acquisition of the Mandarin belief verb *yǐwéi*

Dingyi Pan, Rachel Dudley
Department of Linguistics
{dipan, radudley}@ucsd.edu

Background

Mandarin *yǐwéi*

- Commonly used to indicate someone's belief is false

约翰**以为**(*yǐwéi*)外面在下雨。
John **falsely believes** it is raining outside.
⇝ It is not raining.

约翰**觉得**(*juéde*)外面在下雨。
John **thinks** it is raining outside.
⇝ It might be raining or it might be not.

- Can be used to signal the speaker's belief was false in the past (not morphologically marked)

我**以为**(*yǐwéi*)外面在下雨。
I **falsely believed** it was raining outside.
⇝ It was not raining outside, and I was wrong.

我**觉得**(*juéde*)外面在下雨。
I **think** it is raining outside.
⇝ It might be raining or it might be not.

?

Scarcity of *yǐwéi* in child-directed speech

- *yǐwéi* is rare in child-directed speech.
  Among ~**220,000** child-directed utterances in Mandarin CHILDES: **102** instances of ***yǐwéi*** **1,087** instances of ***juéde*** (to think) **3,213** instances of ***zhīdào*** (to know)

Informativity of *yǐwéi* in theory of mind development

- *yǐwéi* facilitates Mandarin-speaking children's performance on false-belief tasks. [1]
- Children's comprehension ability of *yǐwéi* correlates with their abilities in false-belief task performance. [2]

How do Mandarin-speaking children acquire this false-belief verb *yǐwéi* ?

Potential cues for acquiring belief verbs

- **Syntactic structure** helps constrain the plausible interpretation of a novel word to mentalistic meanings. [3]
- **Contexts** that highlight the salience of mental states, e.g., false-belief scenarios, can further prompt reference to mental states. [4]
  - Contextual information is leveraged in the acquisition of other abstract words. [5,6]

✅ ← Current study with *yǐwéi*

Behavioral studies: HSP with adults

Stimuli

- Child-directed utterances involving the target verbs in dialogues extracted from CHILDES
- 4 conditions x 10 dialogues per condition

Conditions

- Within participants: 2 verbs (*yǐwéi* vs. *juéde*) x 2 types of discourse (constrained vs. unconstrained)
- Between participants: context presence vs. context absence

Context Presence

constrained discourse	unconstrained discourse
ADU: This was really well done. ADU: Who made this? **CHI: (This was) made by Uncle.** ADU: Oh! ADU: It was Uncle who made it? ADU: I ____ it was made by Grandma. ADU: Such a beautiful car. ADU: Oh no!	CHI: Right! ADU: Then who is this? CHI: This is the hen.. ADU: Yeah. ADU: The hen saw that the little sister walked over. ADU: ____ the little sister was going to take the chicks. ADU: And was very angry. ADU: What did the hen do to the little sister?
MOT: Is it a mosquito? CHI: No. MOT: Then what is it? CHI: A ladybug. **MOT: Oh, it is a ladybug.** MOT: You ____ it is a ladybug. MOT: The spider didn't respond. MOT: He's busy spinning the web.	MOT: Look, doesn't it look like an egg? MOT: How to put it together? CHI: Put it together like this! MOT: Put it together like this. MOT: Right. MOT: I also ____ this puzzle is a little hard. MOT: Okay. MOT: Ah, I know.

Information supports ¬*p*: "it was made by grandma" ← *yǐwéi*

Information supports *p*: "it is a ladybug" ← *juéde*

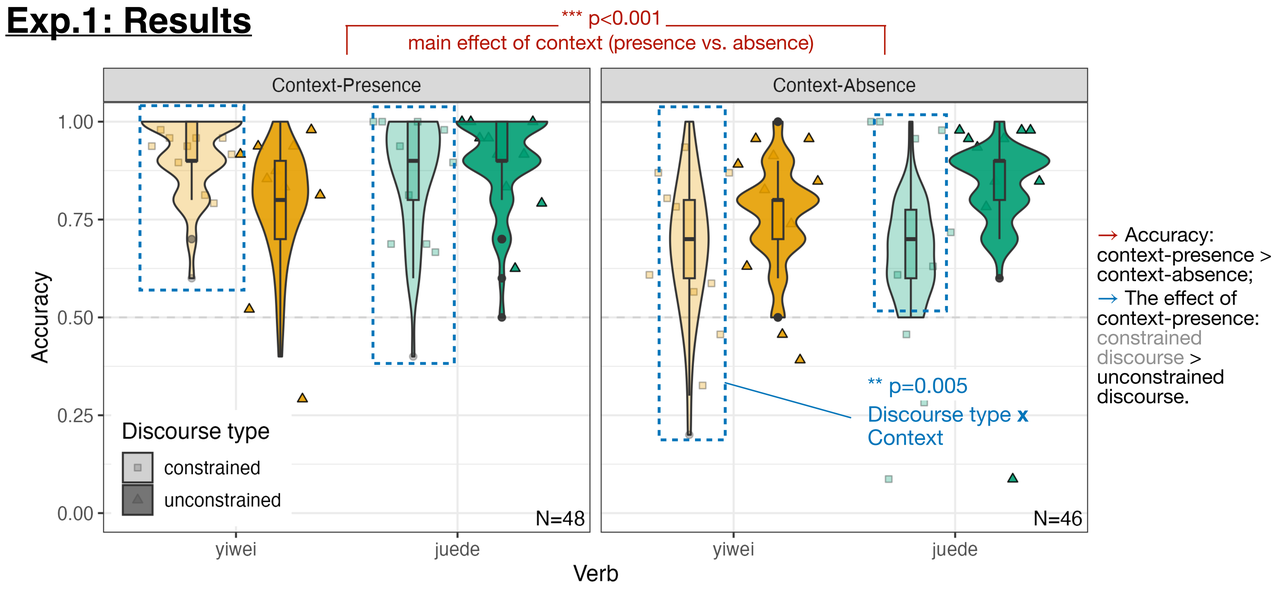
Context Absence

constrained discourse	unconstrained discourse
ADU: I ____ it was made by Grandma.	ADU: ____ the little sister was going to take the chicks.
MOT: You ____ it is a ladybug.	MOT: I also ____ this puzzle is a little hard.

Tasks

- Exp.1: 2AFC, choosing between *yǐwéi* vs. *juéde*
- Exp.2: Open-ended, filling in blank with a verb

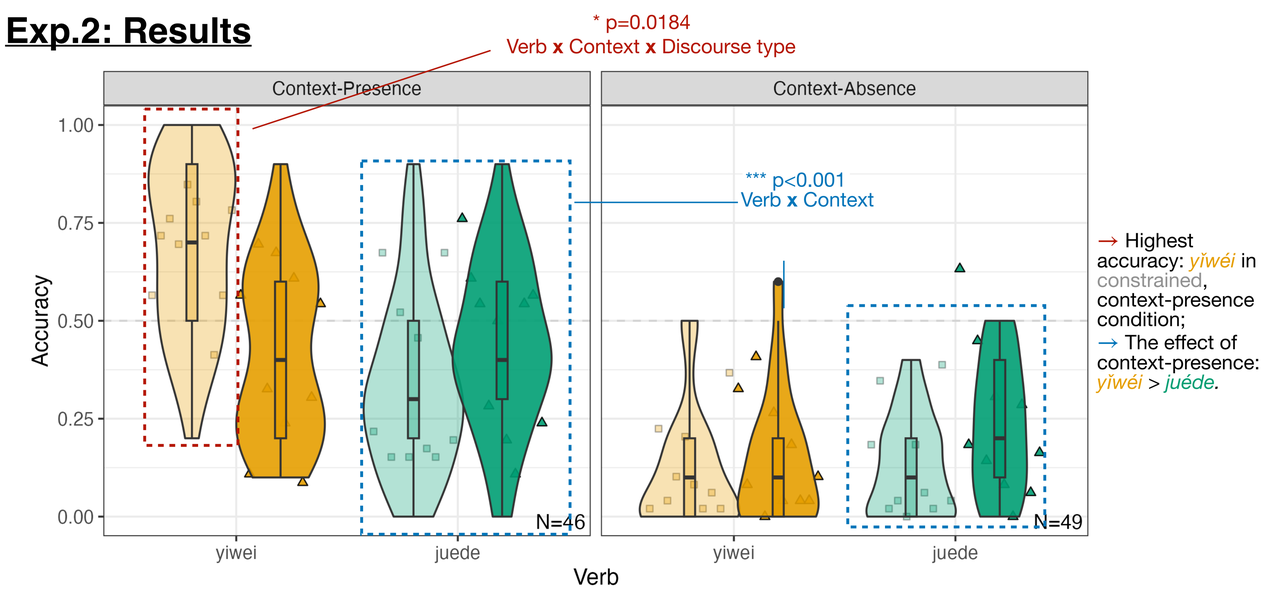
Exp.1: Results



→ Accuracy: context-presence > context-absence;
→ The effect of context-presence: constrained discourse > unconstrained discourse.

➡ High accuracy in context-absence condition: a potential trade off between syntax and discourse cues;
➡ At-ceiling accuracy across all conditions: 2AFC too easy and not necessarily a realistic way to model the learning problem.

Exp.2: Results



→ Highest accuracy: *yǐwéi* in constrained, context-presence condition;
→ The effect of context-presence: *yǐwéi* > *juéde*.

➡ Low accuracy in the context-absence condition: undermines the potential effectiveness of syntactic cues;
➡ Discourse cues are helpful to learn highly specific belief verb *yǐwéi* but not the more general belief verb *juéde.*

Context can serve as a learning cue for this Mandarin false-belief verb *yǐwéi*.

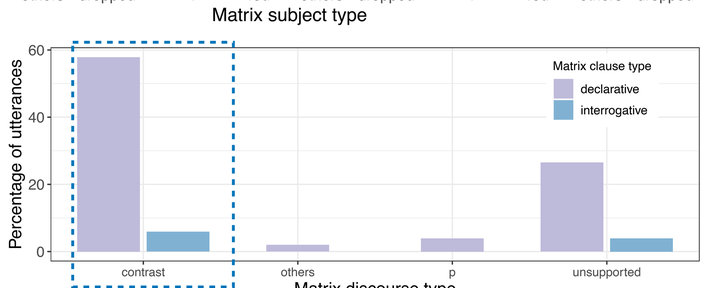
Preliminary corpus results

Syntactic distribution

of *yǐwéi*, *juéde*, and *zhīdào*
Subject type: "*I*","*You*", others (NP, third-person), dropped
Sentence type: declarative, interrogatives

Pragmatic information

of *yǐwéi* only
e.g., for *p* "it is a cat," the type of information in the discourse:
[contrast] That is not a cat./ That is a dog.
[others] Is that a cat?
[p] That is a cat.
[unsupported] Let's play together.



yǐwéi is usually used to express one's own false beliefs and usually occurs in contexts with contrasting information about *p*.

Future directions

- How prevalent is the discourse cue? What are some syntactic cues that might distinguish *yǐwéi* (e.g., adverbial modifications, aspectual and modal markers, etc)?
- When do children start producing *yǐwéi* and do they use it correctly?
- Does *yǐwéi* compete with *juéde* and other belief verbs during acquisition? Is *yǐwéi* harder or easier to learn than *juéde* and other belief verbs? What are some learning cues for *juéde*?

[1] Lee et al. (1999). *Journal of Child Language.* [2] Brandt et al., (2021). *Cognitive Linguistics.* [3] Gillette et al., (1999). *Cognition.* [4] Papafragou et al., (2007). *Cognition.* [5] Gomes et al., (2026). *Cognition.* [6] Sullivan et al. (2019). *Cognitive Science.*